

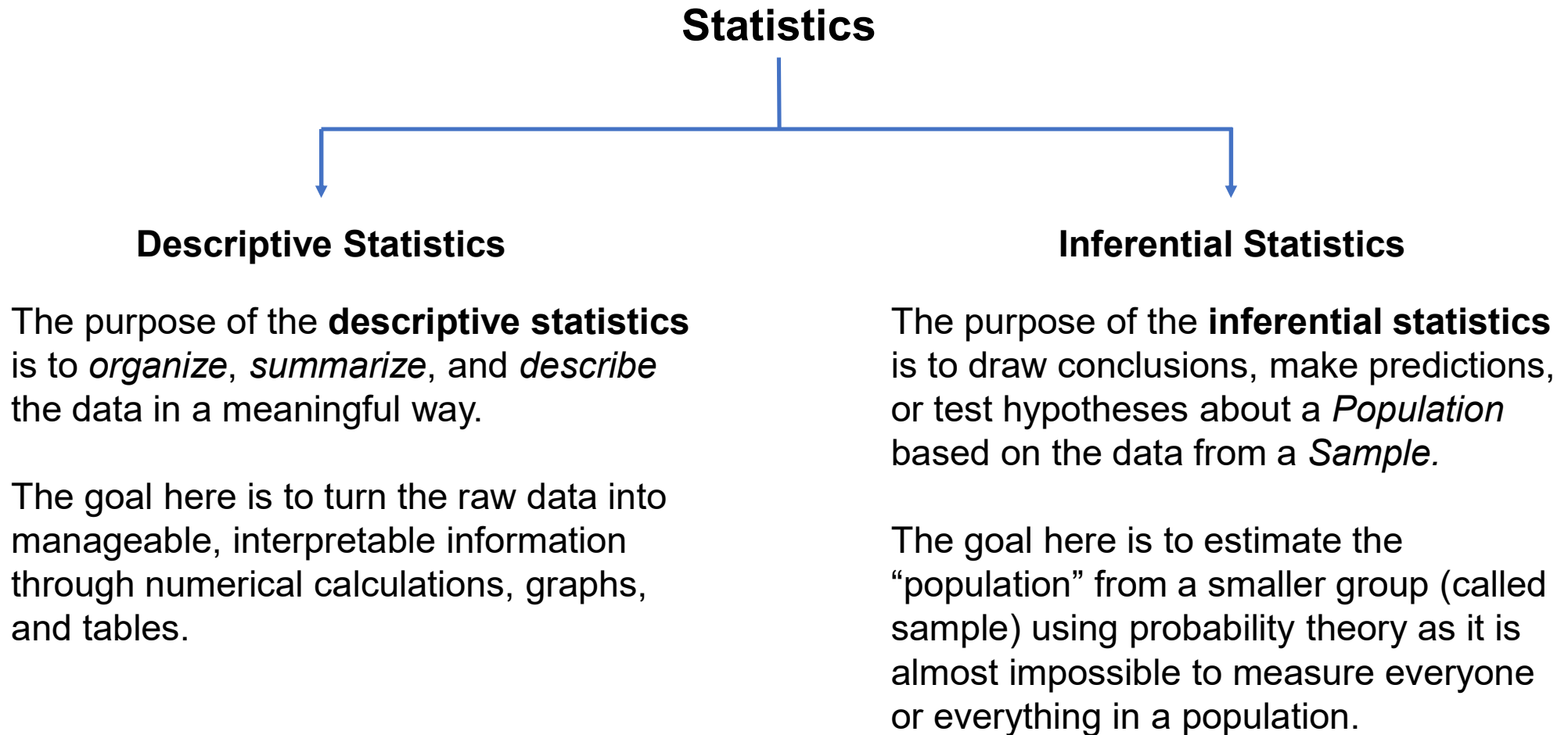
Descriptive Statistics

Sourav Karmakar

souravkarmakar29@gmail.com

Statistics

The subject of “statistics” can be broadly classified into following two disciplines:



Data and Variables

- **Data** are pieces of information about individuals organized into variables.
- By an **individual** (also called **record**), we mean a particular person or object.
- By a **variable**, we mean a particular characteristic of the individual.

The following dataset shows medical records from a particular survey:

Individuals	Variables						
		Gender (M/F)	Age	Weight (lbs.)	Height (in.)	Smoking (1=No, 2=Yes)	Race
	Patient #1	M	59	175	69	1	White
	Patient #2	F	67	140	62	2	Black
	Patient #3	F	73	155	59	1	Asian
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
Patient #75	M	48	90	72	1	White	

Usually variables are arranged across columns while the individuals (records) are arranged across rows

Data and Variables

Variables can be classified into one of two types: categorical or quantitative.

- **Categorical variables** take category or label values and place an individual into one of several groups. Each observation can be placed in *only* one category, and the categories are mutually exclusive.
- **Quantitative variables** take numerical values and represent measurement.

Individuals	Variables						
		Gender (M/F)	Age	Weight (lbs.)	Height (in.)	Smoking (1=No, 2=Yes)	Race
	Patient #1	M	59	175	69	1	White
	Patient #2	F	67	140	62	2	Black
	Patient #3	F	73	155	59	1	Asian
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
	.	.	.	.	.	.	.
Patient #75	M	48	90	72	1	White	

In our example:

- *Gender, Smoking, and Race* are categorical variables.
- *Age, Weight, and Height* are quantitative variables.

Data and Variables

We took a random sample from the 2000 U.S. Census. Here is part of the dataset:

Census is an official count or survey, especially of a population.

US Census 2000

	State	zipcode	Family_Size	Annual_income
1	Florida	32716	8	200
2	Alabama	35236	5	800
3	Florida	32116	6	13500
4	Florida	33679	5	21000
5	Alabama	36374	4	21000
6	California	94565	1	23000

Q.1. Who are the individuals described by this data?

- States
- ✓ People living in the United States in the year 2000
- People with families in the year 2000

The U.S. Census is completed by people living in the United States.

Q.2. What type of variable is Zipcode?

- ✓ Categorical
- Quantitative

Zipcode is a categorical variable because it categorizes individuals by geographic location

Q.3. What type of variable is Annual Income?

- Categorical
- ✓ Quantitative

Annual Income is a Quantitative variable because it assumes continuous values over a range and has arithmetic significance.

Nominal and Ordinal (Categorical) Variables

Nominal Variables

- The word “Nominal” comes from the Latin nomen, meaning “name”. These variables are used to name or label a series of values.
- **Key Features:** There is no natural order or ranking between the categories. One is not “higher” or “better” than the other.
- **Examples:**
 - Marital Status: Single, Married, Divorced, Widowed
 - Blood types: A, B, AB, O
 - Binary/ Dichotomous: Yes/No , Male/Female, Smoker/Non-Smoker *etc.*

Ordinal Variables

- Ordinal variables has “order”
- **Key Features:** The categories have natural, logical order or rank, but the “distance” between the categories is not necessarily measurable.
- **Examples:**
 - Education level: High school, Bachelor’s, Master’s, PhD
 - Ratings of a product: *, **, ***, ****, *****
 - Economic status: Low income, Middle Income, High Income *etc.*

Interval and Ratio (Quantitative) Variables

Interval Variables (or Interval Data)

The “intervals” between the numbers are meaningful, but there is no true zero point. Zero is just another number on the scale; it doesn’t mean “nothing” or “absence”.

- **The “Zero” Rule:** Zero is arbitrary. For example, 0 deg C doesn’t mean there is “no temperature”.
- **“Ratio” test:** You can’t say one value is “twice as much” another. 40 deg C is not “twice as hot” as 20 deg C.
- **Examples:** Temperature, Dates/ Years/ Time of the day, Credit Score, IQ Score *etc.*

Ratio Variables (or Ratio Data)

Ratio data has all the properties of interval data, but it also has a “True Zero”. Zero actually means absence.

- **The “Zero” Rule:** Zero is absolute starting point. 0 kg means no weight.
- **“Ratio” test:** The person who is 200 cm tall is exactly twice as tall as someone who is 100 cm.
- **Examples:** Physical measurements (height, weight, length *etc.*), Time duration, Money, Age *etc.*

Distribution of Variables

- We will begin our journey of Descriptive Statistics by exploring (or looking at) one variable at a time.
- As we saw in the previous slides, usually a variable in a data consist of long list of values (whether numerical or not) and are not very informative in that form.
- In order to convert these raw data into useful information we need to summarize and then examine the **distribution** of the variable. By **distribution** of a variable, we mean:
 - what values the variable takes, and
 - how often the variable takes those values.
- We will first learn how to summarize and examine the distribution of a single categorical variable, and then do the same for a quantitative variable.

Distribution of Variables

Examining Distribution of Categorical Variable

- Suppose we have taken a survey among the office-goers on the mode of daily commutes.
- Few office-going people were randomly chosen and they were asked how they travel from home to office on a regular basis. The dataset we obtain looks something like below.

Persons	Types of Daily Commute
1	Public Transport
2	Public Transport
3	Two - Wheeler
.	.
.	.
.	.
1198	Four - Wheeler
1199	Public Transport
1200	Two - Wheeler

Following are the findings from the given dataset

- There are three mode of transports that people avail.
 - Public Transport
 - Own Two-Wheeler vehicle
 - Own Four-Wheeler vehicle
- There are total 1200 rows or individuals in the dataset

Distribution of Categorical Variable

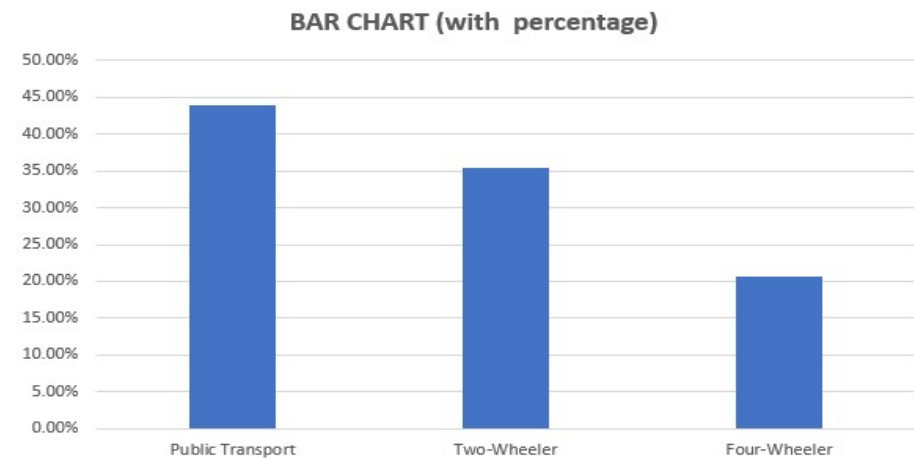
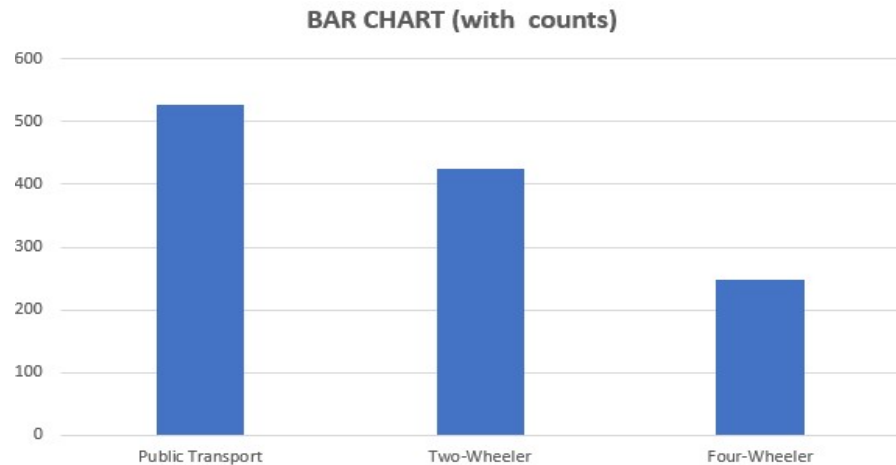
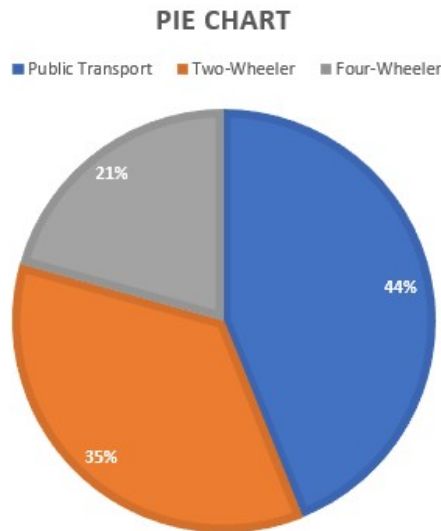
- Suppose we ask: *What percentage of sampled office-goers fall into each category?*
- This question will be easily answered once we summarize and look at the **distribution** of the variable “*Types of Daily Commute*”.
- In order to summarize the distribution of a categorical variable, we first create a table of the different values (**categories**) the variable takes, how many times each value occurs (**count**) and, more importantly, how often each value occurs (by converting the counts to **percentages**); this table is called a frequency distribution. Here is the frequency distribution for our example:

Categories	Count	Percentage
Public Transport	527	43.92%
Two-Wheeler	425	35.42%
Four-Wheeler	248	20.67%
<i>Total</i>	1200	100.00%

Distribution of Categorical Variable

We can express the result pictorially with the help of Bar charts or Pie Charts

Categories	Count	Percentage
Public Transport	527	43.92%
Two-Wheeler	425	35.42%
Four-Wheeler	248	20.67%
<i>Total</i>	1200	100.00%



Distribution of Variables

Examining Distributions of Quantitative Variable

Histogram: Intervals

Break the range of values into intervals and count how many observations fall into each interval.

Examples: Here are the exam grades of 15 students-

88, 48, 60, 51, 57, 85, 69, 75, 97, 72, 71, 79, 65, 63, 73

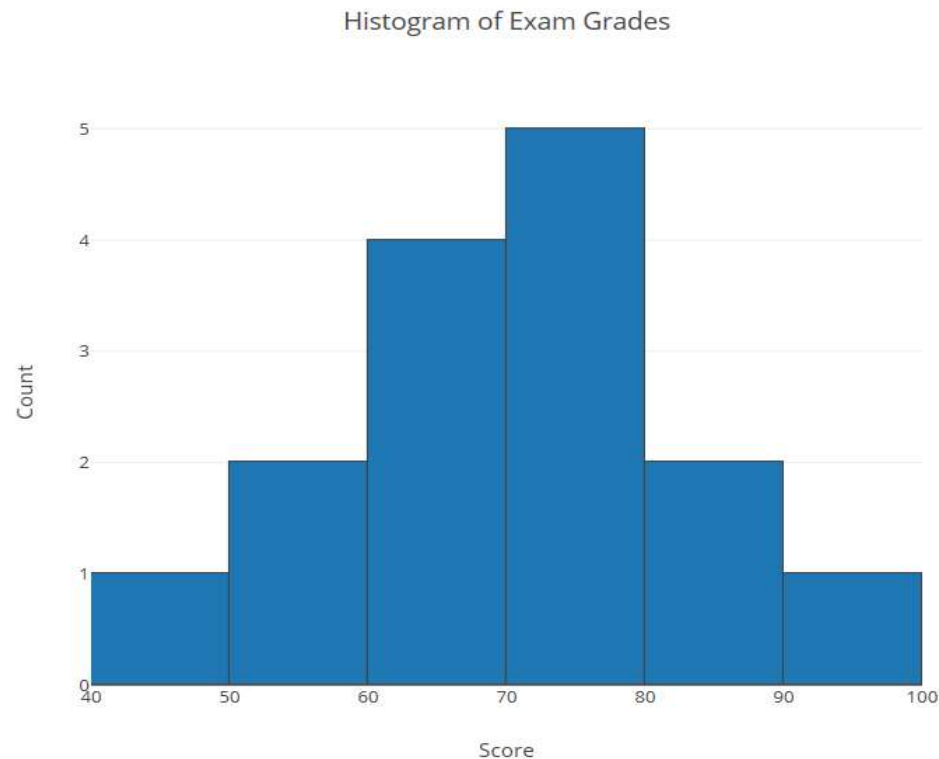
We first need to break the range of values into intervals (also called "bins" or "classes"). By counting how many of the 15 observations fall in each of the intervals, we get the table as shown:

Score Intervals	Counts
[40-50)	1
[50-60)	2
[60-70)	4
[70-80)	5
[80-90)	2
[90-100]	1

Distribution of Quantitative Variable

Score Intervals	Counts
[40-50)	1
[50-60)	2
[60-70)	4
[70-80)	5
[80-90)	2
[90-100]	1

To construct the histogram from this table we plot the intervals on the X-axis and show the number of observations in each interval (frequency of the interval) on the Y-axis, which is represented by the height of a rectangle located above the interval.



Distribution of Quantitative Variable

Measure of Central Tendency:

The three main numerical measures for the center of a distribution are the **mode**, the **mean** and the **median**.

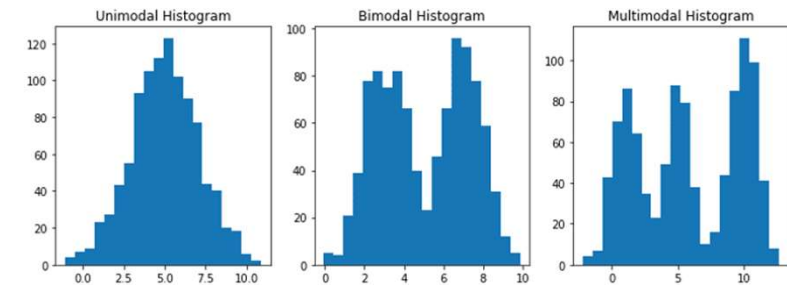
Mode:

The mode is the most commonly occurring value in a distribution.

Mean:

The mean is the average of a set of observations (i.e., the sum of the observations divided by the number of observations). If the n observations are $x_1, x_2, x_3, \dots, x_n$ then their mean or average which we call $\bar{x}$ is calculated as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$



Distribution of Quantitative Variable

Median:

The median is the midpoint of the distribution. It is the number such that half of the observations fall above, and half fall below.

Calculation of Median:

- Order the data from smallest to largest.
- Consider whether n , the number of observations, is even or odd.
 - If n is **odd**, the median M is the center observation in the ordered list. This observation is the one "sitting" in the $\frac{(n+1)}{2}$ **spot** in the ordered list.
 - If n is **even**, the median M is the **mean** of the **two center observations** in the ordered list. These two observations are the ones "sitting" in the $\frac{n}{2}$ and $(\frac{n}{2} + 1)$ spots in the ordered list.

Distribution of Quantitative Variable

Calculation of Median (Example):

Consider the following numbers:

12, 18, 11, 4, 14, 18, 11, 5, 12, 7, 3, 12, 23, 9, 16, 1, 5, 17, 3, 13, 21

- The ordered numbers from small to large is:

1, 3, 3, 4, 5, 5, 7, 9, 11, 11, 12, 12, 13, 14, 16, 17, 18, 18, 21, 23

- Here $n = 21$ (odd). Therefore, the $\frac{(n+1)}{2}$ *th* data i.e. 11th data here will be the median.
Hence, median = 12.

Distribution of Quantitative Variable

Measure of Dispersion / Spread:

These measures provide different ways to quantify the variability of the distribution. We will discuss the following three most commonly used measures of spread.

- Range
- Variance & Standard Deviation and
- Inter-Quartile Range (IQR)

Range:

Range is exactly the distance between the smallest data point (min) and the largest one (Max). i.e.,
 $\text{Range} = \text{Max} - \text{Min}$

Example-1: 12, 18, 11, 4, 14, 18, 11, 5, 12, 7, 3, 12, 23, 9, 16, 1, 5, 17, 3,

Here Max = 23 and Min = 1.

Hence, Range = Max – Min = 23 - 1 = 22

Distribution of Quantitative Variable

Variance & Standard Deviation:

If the n observations are $x_1, x_2, \dots, x_n$ their mean or average is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Then the variance and the standard deviation is calculated as:

$$\text{Variance (Var)} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\text{and Standard Deviation } (\sigma) = \sqrt{\text{Var}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Distribution of Quantitative Variable

Example of calculation of Variance and Standard Deviation:

Consider the following example Dataset:

Sl. No.	Values
1	12
2	5
3	10
4	8
5	5
6	11
7	7
8	9
9	15
10	6

- From the dataset we can observe that number of observations = 10
- Average / Mean =
$$\frac{12+5+10+8+5+11+7+9+15+6}{10} = 8.8$$
- Build the table as shown next.

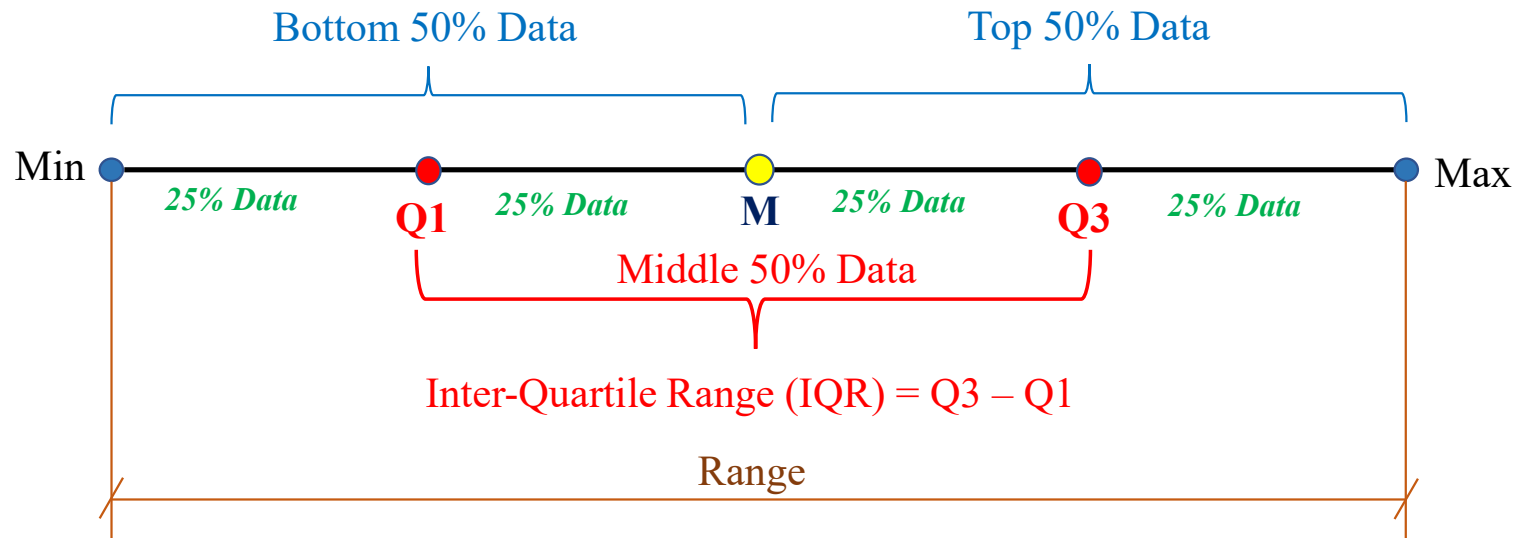
Sl. No.	Values	Values - Mean	(Values - Mean) ²
1	12	3.2	10.24
2	5	-3.8	14.44
3	10	1.2	1.44
4	8	-0.8	0.64
5	5	-3.8	14.44
6	11	2.2	4.84
7	7	-1.8	3.24
8	9	0.2	0.04
9	15	6.2	38.44
10	6	-2.8	7.84

- From the last table, $Variance (Var) = \frac{1}{10} \sum (Values - Mean)^2 = \frac{95.6}{10} = 9.56$
- Hence, $Standard Deviation (\sigma) = \sqrt{9.56} = 3.092$

Distribution of Quantitative Variable

Inter Quartile Range (IQR):

The IQR measures the variability of a distribution by giving us the range covered by the *MIDDLE* 50% of the data. The following figure illustrates the idea.



M: Median of the data (also called second quartile **Q2**)

Q1: First Quartile of the data (one quarter of the data points fall below it)

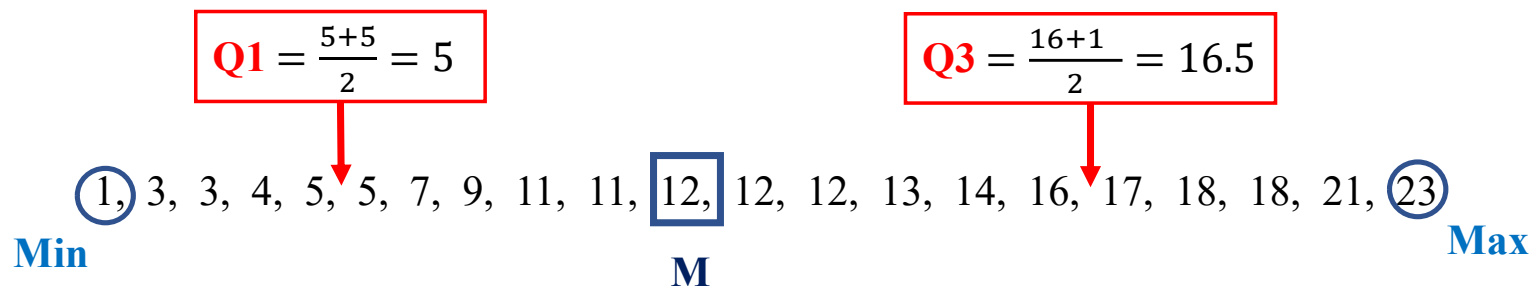
Q3: Third Quartile of the data (three quarters of the data points fall below it, and one quarter above it)

Distribution of Quantitative Variable

Example of Calculating Inter Quartile Range (IQR):

Consider the observations: 12, 18, 11, 4, 14, 18, 11, 5, 12, 7, 3, 12, 23, 9, 16, 1, 5, 17, 3, 13, 21

- Sort the observations in ascending order:



- Find the Median (M), followed by First Quartile (Q1) and Third Quartile (Q3)
- Then calculate Inter Quartile Range (IQR) = $Q3 - Q1 = 16.5 - 5 = 11.5$

The combination of all five numbers (min, Q1, M, Q3, Max) is called the **five number summary**, and provides a quick numerical description of both the center and spread of a distribution.

Distribution of Quantitative Variable

Box and Whisker Plot:

This is the way to visualize the distribution of quantitative variable using **Five Number Summary**.



Five Number Summary

- **M:** Median
- **Q1:** First Quartile
- **Q3:** Third Quartile
- **W₁:** Lower Whisker
- **W₂:** Upper Whisker

There are several methods of drawing whiskers. We'll use the $1.5 \times IQR$ criterion, also known as the **Tukey method** for plotting whiskers. If *Max* and *Min* are the maximum and minimum values in the dataset respectively, then:

- **If:** $Min \geq Q1 - 1.5 \times IQR$, **Then:** $W_1 = Min$
Else: W_1 is at minimum value that is $\geq Q1 - 1.5 \times IQR$
- **If:** $Max \leq Q3 + 1.5 \times IQR$, **Then:** $W_2 = Max$
Else: W_2 is at maximum value that is $\leq Q3 + 1.5 \times IQR$

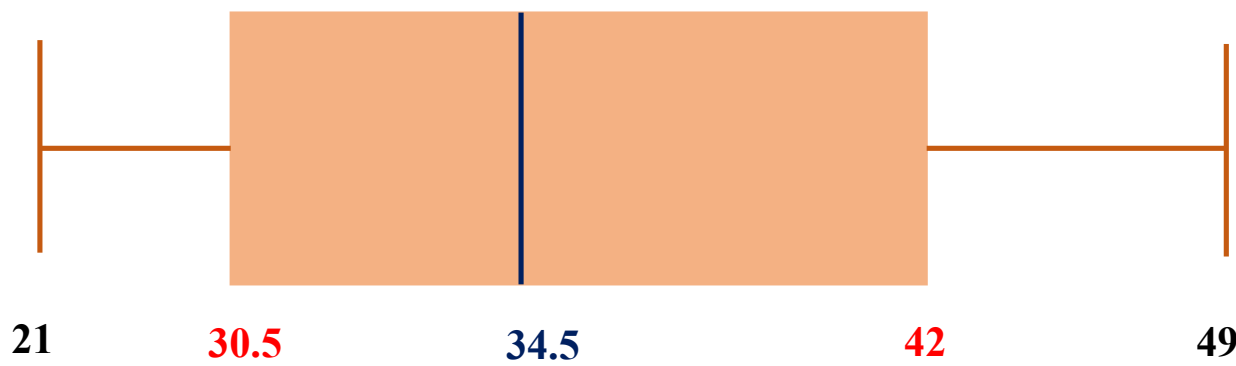
Distribution of Quantitative Variable

Example of Box and Whisker Plot:

Following are the ages of the actresses who won academy award (Oscar) from the year 1970 to 2013:

34, 34, 27, 37, 42, 41, 36, 32, 41, 33, 31, 74, 33, 49, 38, 61, 21, 41, 26, 80, 42, 29, 33, 36, 45, 49, 39, 34, 26, 25, 33, 35, 35, 28, 30, 29, 61, 32, 33, 45, 29, 62, 22, 44

On examining the data we find the following 5 number summary

- Min = 21
 - Q1 = 30.5
 - M = 34.5
 - Q3 = 42
 - Max = 80
- 
- A box and whisker plot representing the distribution of actress ages. The plot features a central orange box with a dark blue vertical line indicating the median at 34.5. The left whisker extends to the minimum value of 21, and the right whisker extends to the maximum value of 49. The box's edges are at the first quartile (30.5) and third quartile (42). The values 21, 30.5, 34.5, 42, and 49 are labeled below the plot line in black, red, blue, red, and black text respectively.
- $IQR = Q3 - Q1 = 42 - 30.5 = 11.5$
 - $Q1 - 1.5 \times IQR = 30.5 - 1.5 \times 11.5 = 13.25$ and $Min > 13.25$. Hence, W_1 is at Min (i.e. at 21)
 - $Q3 + 1.5 \times IQR = 42 + 1.5 \times 11.5 = 59.25$ and $Max > 59.25$. Hence, W_2 is at 49 (which the maximum value that is less than $Q3 + 1.5 \times IQR$)

Outlier detection using Box Plot (Tukey's Method)

Outlier:

In statistics and data science, an **outlier** is a data point that differs significantly from the rest of the observations in a dataset. Think of it as the "odd one out"—a value that lies at an abnormal distance from other values in a random sample.

Example:

For example, if you are measuring the heights of a group of 10-year-olds and most are between 4 and 5 feet tall, but one child is 7 feet tall, that 7-foot measurement is an outlier.

Tukey's method of outlier detection:

In the Box and whisker plot, any value that falls above "Upper Whisker" or falls below "Lower Whisker" is an outlier.

Like, in the previous example the age of actress above 49 or below 21 are outliers. (80, 74, 62, 61)



Deciles and Percentiles

While **Quartiles** divide data into 4 part, **Deciles** and **Percentiles** follow the same logic but offer much finer “slices” of data. Both are positional measures used to describe where a specific value falls within a sorted dataset.

- Deciles (D): Divide the data into **10 equal parts** (each part contains 10% of the data). There are 9 deciles (D_1 to D_9).
- Percentiles (P): Divide the data into **100 equal parts** (each part contains 1% of the data). There are 99 percentiles (P_1 to P_{99}).

The relationship Map

Measures	Equivalent To	Significance
Median	$Q_2 / D_5 / P_{50}$	The exact “center” of the data
First Quartile (Q_1)	P_{25}	The “bottom 25%” boundary
Third Quartile (Q_3)	P_{75}	The “top 25%” boundary
First Decile (D_1)	P_{10}	The “bottom 10%” boundary
Ninth Decile (D_9)	P_{90}	The “top 10%” boundary

Relationships among Variables

While it is important to know how to describe the distribution of a single variable, most studies pose questions that involve exploring the relationship between **two** variables using the collected data. Here are a few examples of such questions with the two variables highlighted:

1. How is the ^Y**number of calories** in a sandwich related to the ^X**type of sandwich** (veg, egg or chicken)?
2. Are the ^Y**smoking habits** of a person (yes/no) related to the person's ^X**gender** (male/female)?
3. What is the relationship between a person's ^X**age** and ^Y**farsightedness**?
4. Is there a relationship between ^X**gender** and ^Y**test scores** on a particular standardized test?
5. Can you predict a person's ^Y**favorite type of music** (classical, rock, jazz) based on his/her ^X**Age**?

In most studies involving two variables, each of the variables has a role. We distinguish between:

- The **explanatory** variable (also commonly referred to as the **independent variable**) — the variable that claims to explain, predict or affect the response. Typically, it is denoted by **X**.
- The **response** variable (also commonly referred to as the **dependent variable**) — the outcome of the study. Denoted by **Y**.

Can you predict the **X** and **Y** of above examples?

Relationships among Variables

If we further classify each of the two relevant variables according to **type** (categorical or quantitative), we get the following 4 possibilities for “**role-type classification**”

- Categorical explanatory and quantitative response ($C \rightarrow Q$)
- Categorical explanatory and categorical response ($C \rightarrow C$)
- Quantitative explanatory and quantitative response ($Q \rightarrow Q$)
- Quantitative explanatory and categorical response ($Q \rightarrow C$)

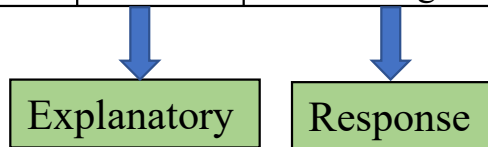
		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

1. How is the **number of calories** in a sandwich related to (or affected by) the **type of sandwich** (veg, egg or chicken) **Ans: $C \rightarrow Q$**
2. Are the **smoking habits** of a person (yes/no) related to the person's **gender** (male/female)? **Ans: $C \rightarrow C$**
3. What is the relationship between a person's **age** and **farsightedness**? **Ans: $Q \rightarrow Q$**
4. Is there a relationship between **gender** and **test scores** on a particular standardized test? **Ans: $C \rightarrow Q$**
5. Can you predict a person's **favorite type of music** (classical, rock, jazz) based on his/her **Age**? **Ans: $Q \rightarrow C$**

C → C Relationships

A survey is done among 1200 college students and they were asked how do they feel about their **body-image** (over-weight / under-weight/ about right) . The table of the data looks like following:

Student	Gender	Body-Image
1	M	over-weight
2	M	about-right
3	F	over-weight
.	.	.
.	.	.
.	.	.
1198	F	under-weight
1199	M	about-right
1200	F	over-weight



Suppose we ask questions like:

- Are men and women just as likely to think their weight is about-right?
- Is there a difference between the genders in feelings about body image?

To answer these type of questions we need to examine the relationship between two categorical variables, **gender** and **body image**.

Here **Gender** is **Explanatory** variable and **Body Image** is the **Response** variable

In order to summarize the relationship between two categorical variables, we create a display called a **two-way table**.

C → C Relationships

Two Way Table:

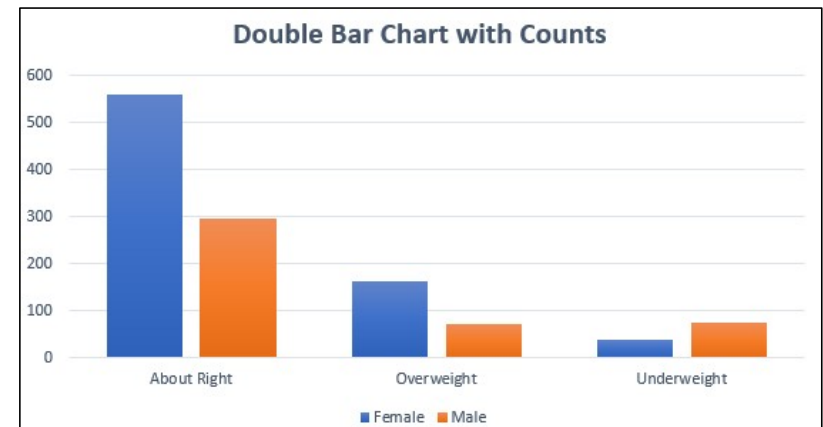
		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

This **Total** gives the summary of the categorical variable **Body Image**

This **Total** gives the summary of the categorical variable **Gender**

A way to visualize the distribution of categorical variables and relationship among them is to plot a **Double Bar Chart**

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200



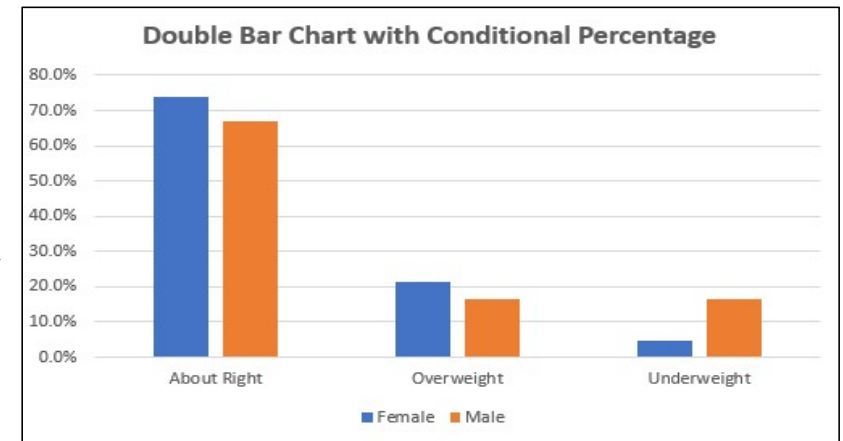
C → C Relationships

As we are examining the relationship among gender and body image, hence we create a conditional percentage table and plot corresponding Double Bar Chart

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200



		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%



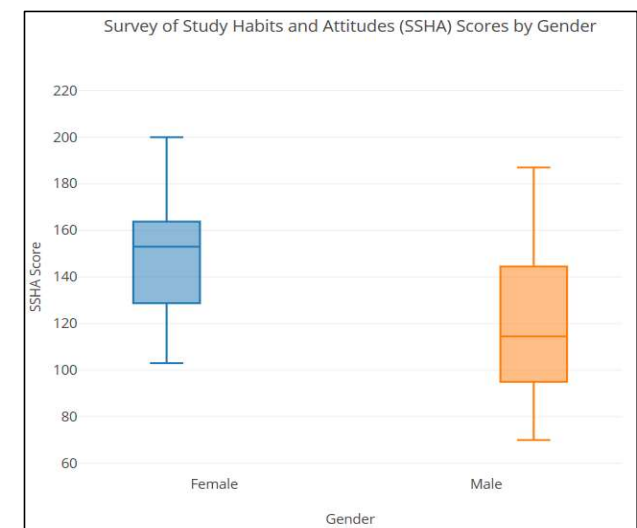
C → Q Relationships

The **Survey of Study Habits and Attitudes (SSHA)** is a psychological test designed to measure the motivation, study habits, and attitudes toward learning of college students. Is there a relationship between **gender** and **SSHA scores**? In other words, is there a "gender effect" on SSHA scores? Data were collected from 40 randomly selected college students, and here is what the raw data look like:

	Explanatory	Response
	Gender	SSHA score
Student 1	Female	154
Student 2	Female	109
Student 3	Male	108
Student 4	Female	115
.	.	.
.	.	.
.	.	.
Student 40	Male	140

Following is the *Five-number summary* of SSHA score separated Gender-wise. Side-by-side boxplots supplemented by descriptive statistics allow us to compare the distribution of SSHA scores within each category of the explanatory variable **gender**:

Statistic	Female	Male
min	103	70
Q1	128.75	95
Median	153	114.5
Q3	163.75	144.5
Max	200	187



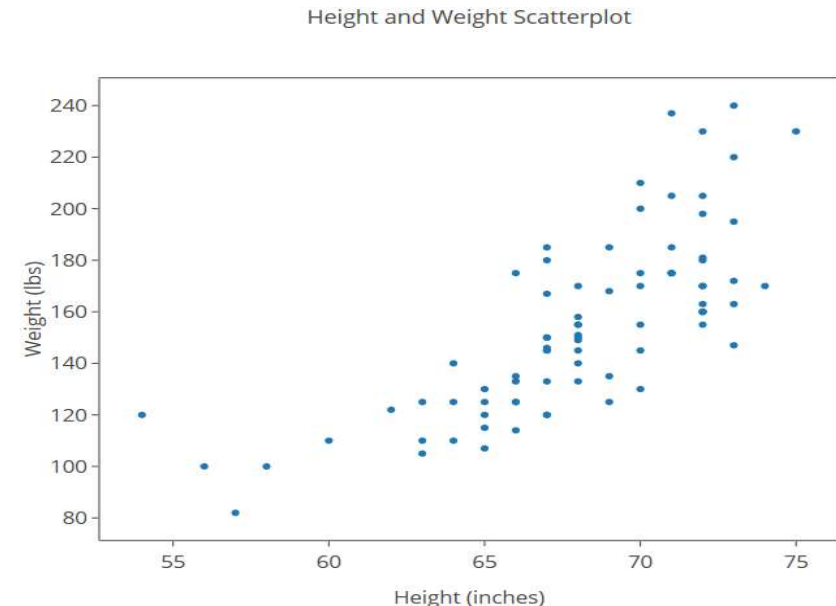
Q → Q Relationships

We look at height and weight data that were collected from 57 males and 24 females and use the data to explore how the weight of a person is related to (or affected by) his or her height. This implies that **height** will be our **explanatory variable** and **weight** will be our **response variable**. The dataset look something like following:

Sl. No.	height (inches)	weight (lbs)	male
1	67	115.3	1
2	62	120	0
3	73.5	142	1
.	.	.	.
.	.	.	.
.	.	.	.
79	71	175	1
80	63	125	0
81	58	100	0

We create scatter plots from the given data which looks like the one shown beside.

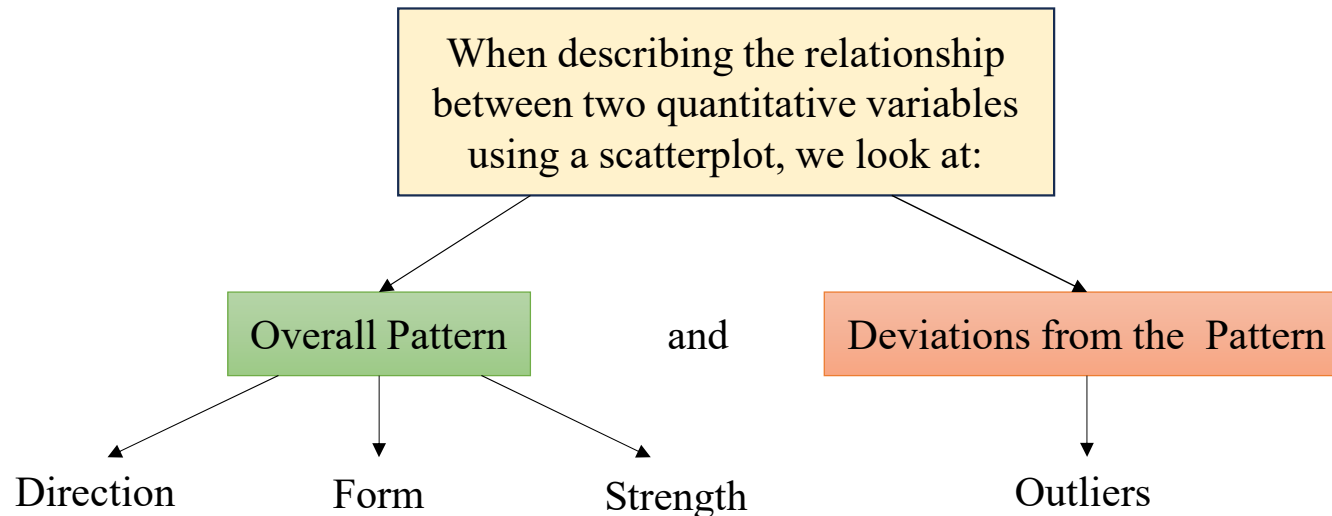
Drawing scatter plot is simple. You just have to plot each points specified by the co-ordinates. Here the co-ordinates are **(height, weight)**



Q → Q Relationships

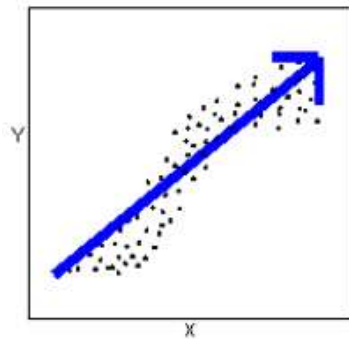
Interpreting the Scatterplot: How do we explore the relationship between two quantitative variables using the scatterplot?

Recall that when we described the distribution of a single quantitative variable with a histogram, we described the overall pattern of the distribution (shape, center, spread) and any deviations from that pattern (outliers). **We do the same thing with the scatterplot.**

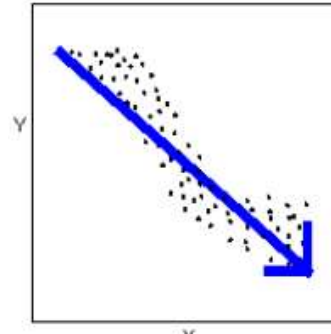


Q → Q Relationships

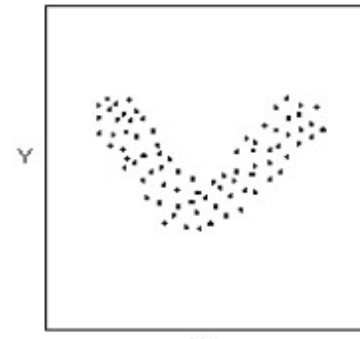
The **direction** of the relationship can be positive, negative, or neither:



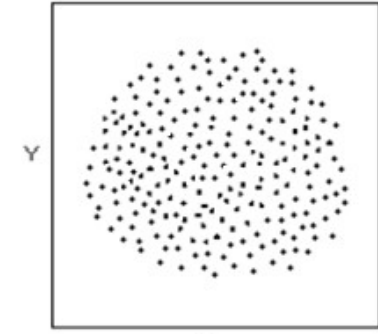
Positive relationship



Negative relationship

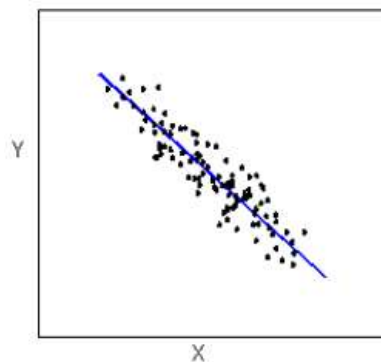


Neither positive
nor negative

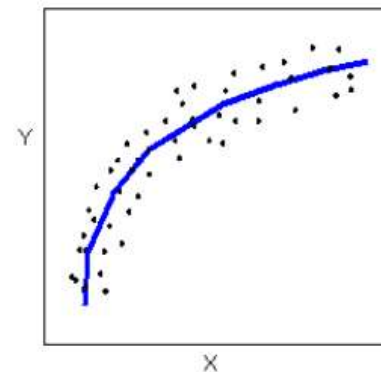


no relationship

The **form** of the relationship is its general shape. When identifying the form, we try to find the simplest way to describe the shape of the scatterplot. There are many possible forms.



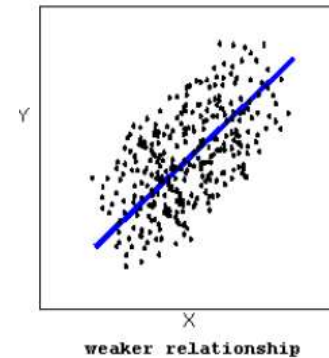
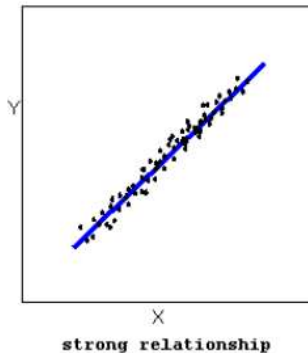
Linear



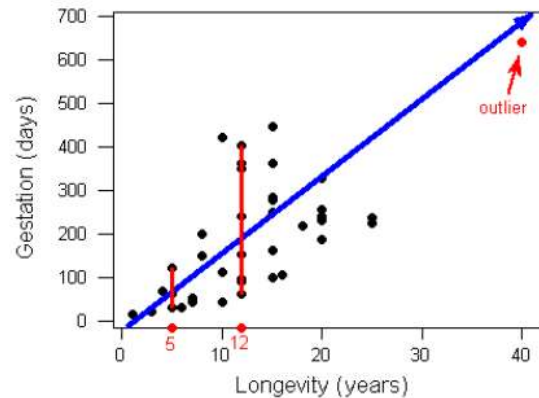
Curvilinear

Q → Q Relationships

The **strength** of the relationship is determined by how closely the data follow the form of the relationship. Let's look, for example, at the following two scatterplots displaying positive, linear relationships:



Consider the following dataset. There is one **outlier** spotted in the dataset.



Can you guess the **direction** of the relationship?

Positive

Can you guess the **form** of the relationship?

Linear

Can you guess the **strength** of the relationship?

Weak

Q → Q Relationships

The Covariance (Cov):

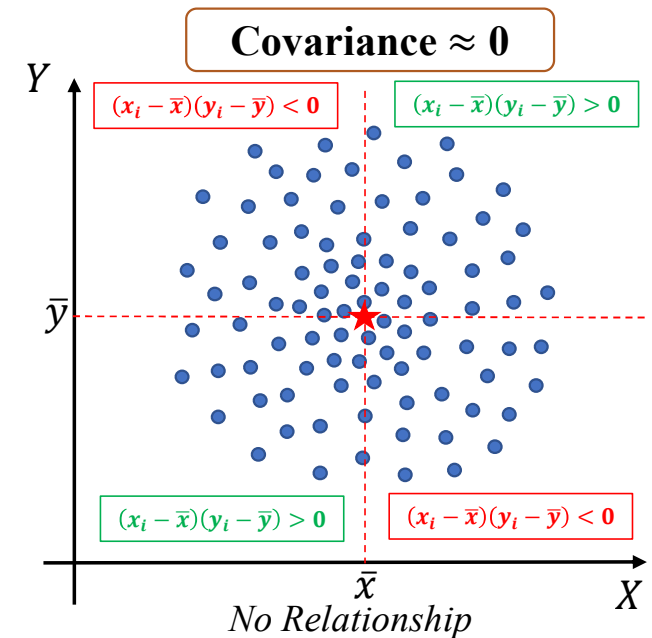
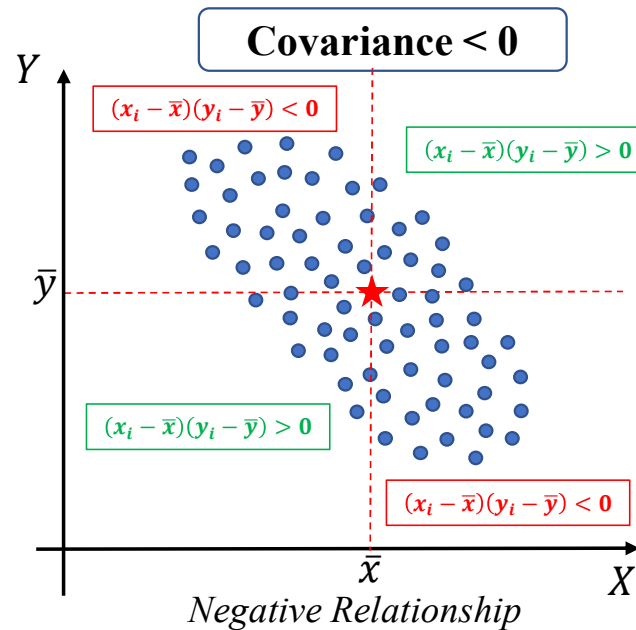
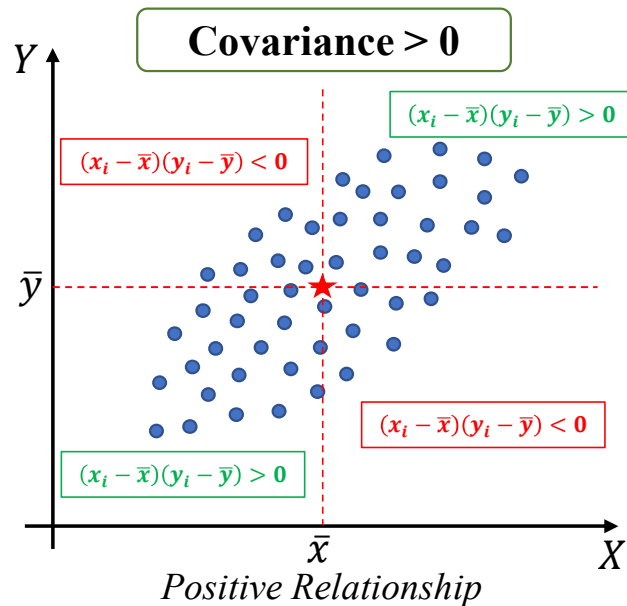
The covariance between two quant. variable measures how variation in one variable affects the variation in another.

Calculation: The covariance between two quantitative variable, X and Y is calculated as following:

X	x_1	x_2	x_3	$\cdot \cdot \cdot$	x_{n-2}	x_{n-1}	x_n
Y	y_1	y_2	y_3	$\cdot \cdot \cdot$	y_{n-2}	y_{n-1}	y_n

$$\text{Covariance}(X, Y) = \text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Intuition of Covariance:



Covariance

X	x_1	x_2	x_3	$\cdot \cdot \cdot$	x_{n-2}	x_{n-1}	x_n
Y	y_1	y_2	y_3	$\cdot \cdot \cdot$	y_{n-2}	y_{n-1}	y_n

Steps for calculating covariance

1. Number of observations = n
2. Compute average of x : $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
3. Compute average of y : $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$
4. Now subtract $\bar{x}$ from all the x_i and $\bar{y}$ from all the y_i
5. Then calculate the covariance using the following formula:

$$\text{Covariance}(X, Y) = \text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Example:

X	Y	$X - \bar{x}$	$Y - \bar{y}$
1	3	-2	-3
2	4	-1	-2
3	5	0	-1
4	7	1	1
5	11	2	5

1. $n = 5$
2. $\bar{x} = \frac{1}{5} (1 + 2 + 3 + 4 + 5) = \frac{15}{5} = 3$
3. $\bar{y} = \frac{1}{5} (3 + 4 + 5 + 7 + 11) = \frac{30}{5} = 6$
4. It is shown in the table above
5. $\text{Cov}(X, Y) = \frac{1}{5} [(-2 \times -3) + (-1 \times -2) + (0 \times -1) + (1 \times 1) + (2 \times 5)] = \frac{19}{5} = 3.8$

Covariance Matrix

If you have multiple quantitative variables, you can compute covariance among different variables and store them in a square matrix called “Covariance Matrix”.

	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>X</i>	$Cov(X, X)$	$Cov(X, Y)$	$Cov(X, Z)$
<i>Y</i>	$Cov(Y, X)$	$Cov(Y, Y)$	$Cov(Y, Z)$
<i>Z</i>	$Cov(Z, X)$	$Cov(Z, Y)$	$Cov(Z, Z)$



	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>X</i>	$Var(X)$	$Cov(X, Y)$	$Cov(X, Z)$
<i>Y</i>	$Cov(Y, X)$	$Var(Y)$	$Cov(Y, Z)$
<i>Z</i>	$Cov(Z, X)$	$Cov(Z, Y)$	$Var(Z)$

Now, covariance of a variable with itself is nothing but variance of that variable.

$$Cov(X, X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = Var(X)$$

Now, also notice that.

$$Cov(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = Cov(Y, X)$$

Hence, Covariance matrix is a symmetric matrix.

Q → Q Relationships

The Correlation Coefficient (r):

The numerical measure that assesses the strength of a linear relationship is called the **correlation coefficient**.

Definition: The **Pearson's correlation coefficient (r)** is a numerical measure that measures the **strength** and **direction** of a *linear relationship* between two quantitative variables.

Calculation: The **Pearson's correlation coefficient (r)** between two quantitative variable X and Y is calculated as

X	Y
x_1	y_1
x_2	y_2
x_3	y_3
$\cdot$	$\cdot$
$\cdot$	$\cdot$
$\cdot$	$\cdot$
x_{n-2}	y_{n-2}
x_{n-1}	y_{n-1}
x_n	y_n

Let σ_x and σ_y be the standard deviations of X and Y respectively and there are n data-points. Then the correlation coefficient between X and Y (denoted by r_{xy} is):

$$r_{xy} = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right)$$

Expanding σ_x and σ_y and noting that $\bar{x}$ and $\bar{y}$ are the mean of X and Y respectively. Then the correlation coefficient between X and Y (denoted by r_{xy} is) can be written as:

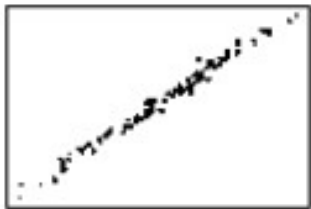
$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{Cov(X, Y)}{\sqrt{Var(X)} \sqrt{Var(Y)}}$$

Q → Q Relationships

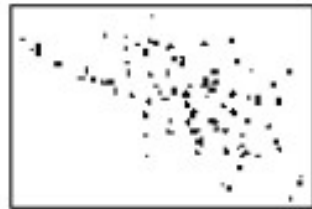
Interpreting the value of Correlation Coefficient (r):

The **correlation coefficient** lies between -1 to +1, i.e., $|r| \leq 1$ or $-1 \leq r \leq 1$

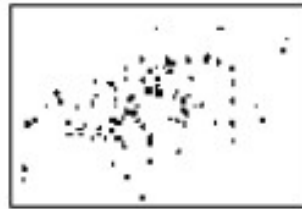
- Positive value of r indicates positive relationship.
- Negative value of r indicates negative relationship.
- r value close to 1 indicates strong positive linear relationship.
- r value close to -1 indicates strong negative linear relationship.
- r value close to 0 indicates the relationship is neither positive nor negative.



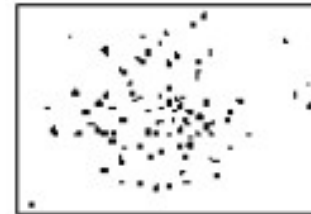
$r = 0.995$



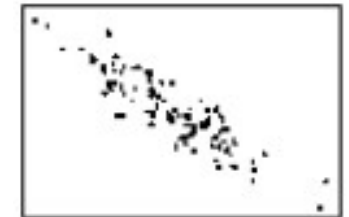
$r = -0.575$



$r = 0.436$



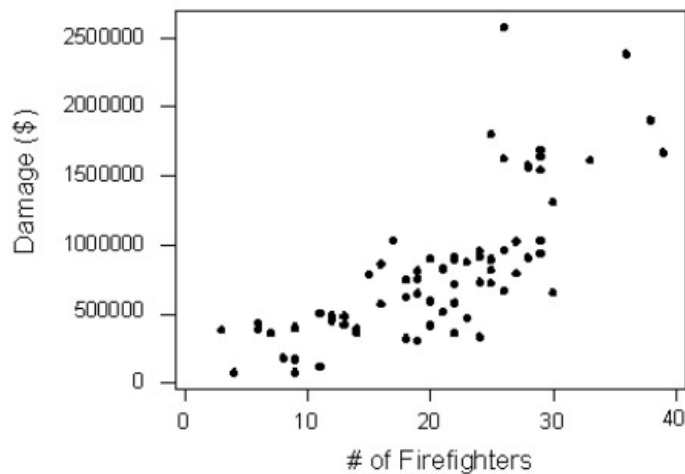
$r = 0.100$



$r = -0.897$

Causation

The scatterplot below illustrates how the number of firefighters sent to fires (X) is related to the amount of damage caused by fires (Y) in a certain city.



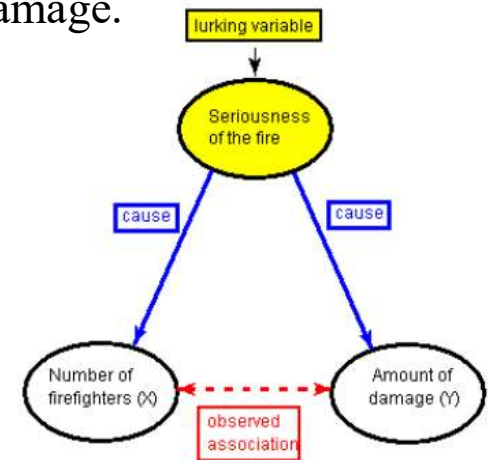
The scatterplot clearly displays a (slightly curved) **positive** relationship between the two variables. Would it, then, be reasonable to conclude that sending more firefighters to a fire causes more damage, or that the city should send fewer firefighters to a fire, in order to decrease the amount of damage done by the fire?

Of course not! So, what is going on here?

There is a **third variable in the background**—the seriousness of the fire—that is responsible for the observed relationship. More serious fires require more firefighters and cause more damage.

Here, the seriousness of the fire is a **lurking variable**. A **lurking variable** is a variable that is not among the explanatory or response variables in a study but could substantially affect your interpretation of the relationship among those variables.

Hence, Correlation doesn't imply Causation



Thank You